Simulation of NO_x Emission in Circulating Fluidized Beds Burning Low-grade Fuels[†]

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Nitrogen oxides are a major environmental pollutant resulting from combustion. This paper presents a modeling study of pollutant NO_x emission resulting from low-grade fuel combustion in a circulating fluidized bed. The simulation model accounts for the axial and radial distribution of NO_x emission in a circulating fluidized bed (CFB). The model results are compared with and validated against experimental data both for small-size and industrial-size CFBs that use different types of low-grade fuels given in the literature. The present study proves that CFB combustion demonstrated by both experimental data and model predictions produces low and acceptable levels of NO_x emissions resulting from the combustion of low-grade fuels. Developed model can also investigate the effects of different operational parameters on overall NO_x emission. As a result of this investigation, both experimental data and model predictions show that NO_x emission increases with the bed temperature but decreases with excess air if other parameters are kept unchanged.

Introduction

Circulating fluidized bed (CFB) combustion is receiving wide research attention in view of its potential as an economic and environmentally acceptable technology for burning low-grade coals. In addition to highly efficient operation, a combustion system should comply with the requirement of minimizing environmental impact. The emission rate of various pollutants from the combustion of coal depends on fuel analysis, combustor design, and operating conditions. Fluidized bed combustion allows clean and efficient combustion of coal. Designing of the CFB combustor (CFBC) is very important because of burning coal with high efficiency and within acceptable levels of gaseous emissions. A good understanding of the combustion and pollutant generating processes in the combustor can greatly avoid costly upsets. One of the major advantages of CFBCs is their efficiency for combustion of low-grade lignites.[1,2]

Nitrogen oxides are a major environmental pollutant resulting from combustion. The reactions of nitrogen oxides with carbons or chars are of current interest with regard to their possible role in reducing NO_x emissions from combustion systems. They also offer new useful insights into the oxidation reactions of carbons, generally.[3] A large amount of literature concerning these reactions has developed, as evidenced in three reviews[4−6] and by the recent publication of many papers in the area.[1,7,8] These works have suggested considerable complexity in the mechanisms of NO_x reduction and a large variability in reported kinetics. There are two approaches to describe NO_x emission in CFB.[9] The first approach involves overall reaction (considering catalytic activity of CaO and char). The overall rate constants are measured preferably under CFB conditions.[10] The other approach is more thorough and is based on actual chemical reactions whose rate constants can be taken from literature.[11] For CFB only 106 reactions with 28 species were used to model the NO_x emission. However, a detailed review shows that all N-related reactions have not the same importance.[12] So instead of considering all N-related reactions, one could use only the important reactions for the development of a predictive procedure for the overall NO_x emission from a CFBC.

The objective of the model presented in this study is to be able to predict the pollutant emissions formation and destruction of different low-grade Turkish lignites in various sizes of CFBCs. There are considerable reserves of lignite in Turkey. Most of Turkish lignite reserves are of low-grade lignites with a calorific value of about 12 000 kJ/kg, ash content of about 25−30%, and average sulfur content of about <4%. The main problem for Turkish units running on lignite is presented by the air emissions.[13]

For the reduction of pollutant emissions from coal fired power plants, numerous techniques, involving the staged input of fuel and air, have been successfully applied. The application of these techniques to industrial-scale combustors necessitates combus-

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tion parameters optimization that is extremely time-consuming and expensive. Mathematical modeling allows the testing of many variable combustion parameters in a much shorter time period and at lower costs. Therefore, mathematical modeling application in the CFB combustion process to enhance combustion performance and to reduce pollutants is seen as an attractive solution. This paper presents a modeling study of NO_x pollutant emission resulting from coal combustion in a CFBC. Using this model, overall NO_x emission is predicted for the combustion of three different kinds of low-grade Turkish lignites. The contents of these lignites are as follows: ash from 23.70 to 45.31%, sulfur from 1.81 to 8.40%, and calorific values (LHV) from 10 283 to 15 215 kJ/kg. The data is obtained from two pilot scale CFBCs (50 and 80 kW) and an industrial scale CFBC (160 MW). The developed model can also investigate the effects of different operational parameters on overall NO_x emission.

Model Description. On the basis of previous work on dynamic two-dimensional (2D) coal combustion modeling of CFBCs,[8] a modeling study of pollutant emissions resulting from coal combustion in CFBCs is present in this study. The present CFBC model can be divided into three major parts: a submodel of the gas−solid flow structure; a reaction kinetic model for local combustion; and a convection/dispersion model with reaction. The latter formulates the mass balances for the gaseous species and the char at each control volume in the flow domain. Kinetic information for the reactions is supplied by the reaction kinetic submodel, which contains a description of devolatilization and char combustion, and emission formation and destruction, respectively.

Hydrodynamics Structure. Combustor hydrodynamic is modeled taking into account a previous work.[14] The model addressed in this paper uses a particle-based approach that considers 2D motion of single particles through fluids. According to the axial solid volume concentration profile, the riser is axially divided into the bottom zone and the upper zone. The bottom zone in the turbulent fluidization regime is modeled in detail as two-phase flow that is subdivided into a solid-free bubble phase and a solid-laden emulsion phase. A single-phase back-flow cell model is used to represent the solid mixing in the bottom zone. A two-phase model is used for gas phase material balance. In the upper zone core−annulus solids flow structure is established. It is assumed that the particles move upward axially and move from core to the annulus region radially. Thickness of the annulus varies according to the bed height. In the annulus region, the particle has a zero normal velocity. The pressure drop through the bottom zone is equal to the weight of the solids in this region and is considered only in the axial direction. In the upper zone, pressure drop due to the hydrodynamic head of solids is considered in the axial direction while pressure drop due to solids acceleration is also considered in the axial and radial directions. The solids friction and gas friction components of pressure drop are considered as boundary conditions in momentum equations for solid and gas phases, respectively in the model. Solids friction is defined as the frictional force between the solids and the wall, wheres the gas friction is the frictional force between the gas and the wall.

The hydrodynamic model takes into account the axial and radial distribution of voidage and velocity, for gas and solid phase, pressure drop for gas phase, and solids volume fraction and particle size distribution for solid phase. In order to determine the validity of the hydrodynamic model, the model results are compared with test results using the same input variables in the tests as the simulation program input. Hydrodynamic model results are compared with experimental results obtained from various CFB test rigs at different size in the literature, where ranges are as follows: bed diameter from 0.05 to 0.418 m, bed height from 5 to 18 m, mean particle diameter from 67 to 520 μm, particle density from 1398 to 2620 kg/m^3, mass fluxes from 21.3 to 300 kg/m^{2}s, and gas superficial velocities from 2.52 to 9.1 m/s. The model compares with experimental results for void fraction, solids volume fraction, and particle velocity along the bed height and bed radius and different bed operational parameters prove the model hydrodynamic structure validation axially and radially. Additionally, inward solids mass flux along the bed radius and pressure gradient along the bed height is validated.[14] The structure and details of the hydrodynamic model are given in a previous study.[14] As some of the results of the model, the variation in mean particle diameter and superficial velocity does affect the pressure especially in the core region, and it does not affect considerably the pressure in the annulus region. The radial pressure profile is getting flatter in the core region as the mean particle diameter increases. Similar results can be obtained for lower superficial velocities. It has also been found that the contribution to the total pressure drop by gas and solids friction components is negligibly small when compared to the acceleration and solids hydrodynamic head components. At the bottom of the riser, in the core region the acceleration component of the pressure drop in total pressure drop changes from 0.65 to 0.28% from the riser center to the core−annulus interface, respectively; within the annulus region the acceleration component in total pressure drop changes from 0.22 to 0.11% radially from the core−annulus interface to the riser wall. On the other hand, the acceleration component weakens as it moves upward in the riser, decreasing to 1% in both regions at the top of the riser, which is an important indicator of the fact that the hydrodynamic head of solids is the most important factor in the total pressure drop.

Kinetic Model. The combustor model takes into account the devolatilization of coal and subsequent combustion of volatiles followed by residual char. As a result of the experimental studies carried out using various types of Turkish lignite, it is known that volatilization products enter the upper region in fluid beds working at slower rates than CFBs.[15,16] In the model, volatiles are entering the combustor with the fed coal particles. It is assumed that the volatiles are released in emulsion phase in the bottom zone of the CFBC at a rate proportional to the solid mixing rate. The degree of devolatilization and its rate increase with increasing temperature. The composition of the products of devolatilization in weight fractions is estimated from the correlations proposed by Loison and Chauvin.[17]

The bed material in the combustor consists of coal, inert particles, and limestone. The properties and size distribution of particles have significant influence on the hydrodynamics and combustion behavior in the CFBC.[18] The model also considers the particle size distribution due to fuel particle fragmentation,[19] char combustion,[20] and particle attrition.[21] Particles in the model

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Table 1. The NO_x Reactions and Reaction Rates used in the Model[8]

reaction		reaction rate
$HCN + {}^1/_2O_2 \rightarrow CNO$	$k = 2.14 \times 10^5 \exp(-10\,000/T)$	$R_{HCN} = kC_{O_2}C_{HCN}$ (mol/(m^3 s))
$CNO + {}^1/_2O_2 \rightarrow NO + CO$	$(k_2)/(k_1) = 1.02 \times 10^9 \exp(-25\,460/T)$	$R_{CNO-O_2} = kC_{O_2}C_{HCN}(k_1/(k_1 + k_2C_{NO}))$ (mol/(m^3 s))
$CNO + NO \rightarrow N_2O + CO$	$k = 2.14 \times 10^5 \exp(-10\,000/T)$	$R_{CNO-NO} = kC_{O_2}C_{HCN}((k_2C_{NO})/(k_1 + k_2C_{NO}))$ (mol/(m^3 s))
$N_2O + C \rightarrow N_2 + CO$	$k = 2.9 \times 10^9 \exp(-16\,983/T)$	$R_{N_2}OC = kN\pi d_c^2C_{N_2}O$ (mol/s)
$N_2O + CO \rightarrow N_2 + CO_2$	$k = 5.01 \times 10^{13} \exp((-4.40 \times 10^4)/(R_uT))$	$R_{N_2}O-CO = kC_{N_2}OC_{CO}$ (mol/(cm^3 s))
$N_2O + {}^1/_2O_2 \rightarrow N_2 + O_2$	$k = 1.00 \times 10^{14} \exp((-2.80 \times 10^4)/(R_uT))$	$R_{N_2}O-O_2 = kC_{N_2}OC_{O_2}$ (mol/(cm^3 s))
$NO + C \rightarrow {}^1/_2N_2 + CO$	$k = 5.85 \times 10^7 \exp(-12\,000/T)$	$R_{NOC} = kN\pi d_c^2C_{NO}$ (mol/s)
$NO + {}^1/_2C \rightarrow {}^1/_2N_2 + {}^1/_2CO_2$	$k = 1.3 \times 10^5 \exp(-17\,111/T)$	$R_{2NOC} = kN\pi d_c^2C_{NO}$ (mol/s)
$NO + CO \rightarrow {}^1/_2N_2 + CO_2$	$KT = 1.952 \times 10^{10} \exp(-19\,000/T)$	$R_{NOCO} = KT((k_1C_{NO}(k_2C_{CO} + k_3)))/((k_1C_{NO} + k_2C_{CO} + k_3))$ (mol/(m^3 s))
	$k_1 = 0.1826$	
	$k_2 = 0.00786$	
	$k_3 = 0.002531$	
$NH_3 + {}^5/_4O_2 \rightarrow NO + {}^3/_2H_2O$	$k = 2.73 \times 10^{14} \exp(-38\,160/T)$	$R_{NH_3}NO = kC_{NH_3}C_{O_2}$ (mol/(m^3 s))
$NH_3 + {}^3/_4O_2 \rightarrow {}^1/_2N_2 + {}^3/_2H_2O$	$k = 3.38 \times 10^7 \exp(-10\,000/T)$	$R_{NH_3}N_2 = (kC_{NH_3}C_{O_2})/(C_{O_2} + k')$ (mol/(m^3 s))
	$k' = 0.054$	
$NO + NH_3 + {}^1/_2O_2 \rightarrow N_2 + {}^3/_2H_2O$	$k = 1.1 \times 10^{12} \exp(-27\,680/T)$	$R_{NONH_3} = k\sqrt{C_{O_2}}\sqrt{C_{NH_3}}\sqrt{C_{NO}}$ (mol/(m^3 s))

are divided into 10 size groups in the model. The Sauter mean diameter is adopted as average particle size. Particles in the bottom zone include particles coming from the solid feed and recirculated particles from the separator.

Kinetics of char combustion is modeled with a shrinking core with attiring shell, that is, the dual shrinking-core model (assuming that the ash separated once formed) with mixed control by chemical reaction and gas film diffusion. The char combustion at each control volume is considered according to the reaction

$$C + \frac{1}{\Phi}O_2 \rightarrow \left(2 - \frac{2}{\Phi}\right)CO + \left(\frac{2}{\Phi} - 1\right)CO_2 \qquad (1)$$

and the char combustion reaction rate, k_c, is calculated as follows:[20,22]

$$k_c = \frac{R_uT/M_c}{\frac{1}{k_{cr}} + \frac{1}{k_{cd}}}\left(\frac{kg}{s}\right)k_{cr} = 8710\exp\left(\frac{-1.4947 \times 10^8}{R_uT}\right)\left(\frac{kg}{m^2s{\cdot}kPa}\right)$$
$$k_{cd} = \frac{12Sh\Phi D_g}{d_pR_gT}\left(\frac{kg}{m^2s{\cdot}kPa}\right)Sh = \frac{k_gd_p}{D_g} = 2\varepsilon + 0.69\left(\frac{Re_p}{\varepsilon}\right)^{1/2}Sc^{1/3} \qquad (2)$$

In the model, the attrition constant value is taken as 2×10^{-7} for the coal particles in the model calculations in both bottom zone and upper zone and the attrition constant value of the coal ash particles is taken as 1.7×10^{-7} as presented in.[15,16]

For the estimation of combustion efficiency (as percentage of the LHV), the heat losses owing to incomplete combustion (accounting the CO emission) and unburned carbon contained in particular matter are determined.[23] In addition to this definition, the losses due to the flue gases to ambient and due to the bed-to-wall heat transfer are taken into account for the estimation of combustion efficiency in this study. According to the this assumption the combustion efficiency of CFBC based on the losses can be defined as follows:

$$\eta = 1 - \frac{\dot{Q}_{unburnt\ carbon} + \dot{Q}_{ash} + \dot{Q}_{unburnt\ CO} + \dot{Q}_{fluegases} + \dot{Q}_{wall}}{\text{Heat rate of fuel}} \qquad (3)$$

NO_x Emission. It was shown in the literature that[24] rather low NO_x emissions are obtained by staged combustion in a fluidized bed combustor. By the use of primary and secondary air injected at different locations in a circulating fluidized bed combustor, its temperature and combustion atmosphere is well-regulated, and generally low NO_x emissions of about 150−350 ppm are reported.[25]

It is crucial to well evaluate the mechanism of NO_x formation to reduce NO_x in the combustor. However, the mechanism of NO_x formation is complex. NO_x formations in combustion processes result from a combination of a thermal generation process and fuel nitrogen oxidation. At very high temperatures, thermal generation of NO_x from the air nitrogen becomes very important, whereas at low temperatures found in a CFBC, the dominant source of NO_x is fuel-nitrogen oxidation.[4−6] Typically, significant amounts of the fuel-nitrogen remain in the char after the devolatilization. The oxidation of this char-nitrogen gives an important contribution to the total nitrogen oxide emissions from the combustor. The mechanism of char-nitrogen oxidation to the products is very complex, and includes not only several homogeneous and heterogeneous reactions but also mass transfer effects inside the pore system of the char and in the boundary layer surrounding the particle.[25] In the present study, fuel-NO_x can be formed through combustion of the nitrogenous species released with volatile matter (such as HCN, NH_3) and oxidation of the nitrogen retained in the char. These reactions, resulting in rapid formation of NO_x, are most likely to proceed in the bottom zone. Meanwhile, in zones with volume O_2 concentrations lower than 10−12%, the NH_3 concentration is probably elevated due to the rapid formation of NH_3 from HCN[26] as well as because of the emission of NH_3 released with volatiles from fuel particles present in these zones. In the upper zone (with lower O_2 concentrations) this may lead to NO_x reduction through its reaction with NH_3, followed by formation of nitrogen gas and water vapor, that is, neutral products. The alternative mechanisms of NO_x reduction in the upper zone involve reactions of NO_x with carbon and CO on the char surface,[27] which are highly probable when firing high-ash fuels. The chemical reactions with their corresponding reaction rates for NO_x emissions formation and retention in the model are given in Table 1. It must be noted that nitrogen oxides generally originate form fuel-nitrogen in forms of NO, NO_2 and N_2O, whereas the model takes into account NO and N_2O as nitrogen oxides. In this study, only NO is considered in the simulation.

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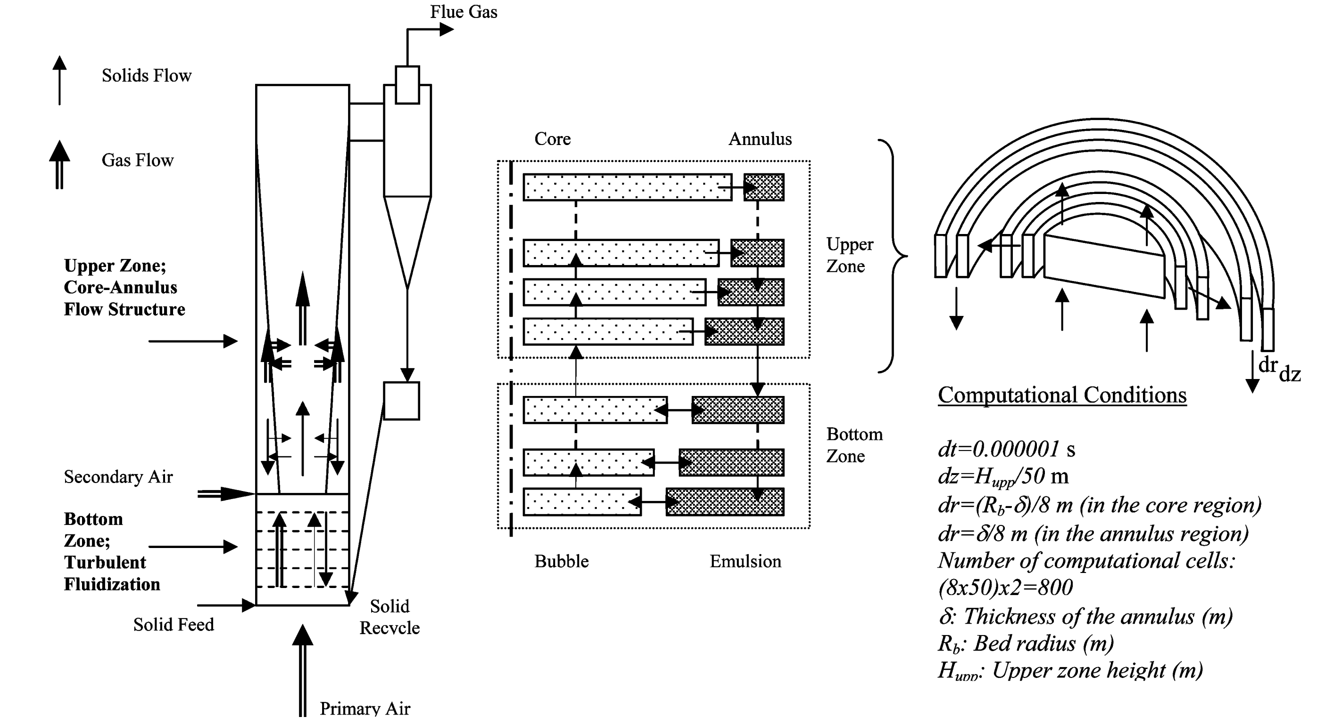

Figure 1. The scheme and the calculation domain of the CFBC.

Table 2. The Conservation of Mass, Momentum, and Energy Equations for Each Phase and the Constitutive Relations

gas phase		solid phase	
Continuity Equation			
$(d(C_j\varepsilon_i))/(dt) = \sum_{in} \dot{n}_j\varepsilon_i - \sum_{out} \dot{n}_j\varepsilon_i + \dot{R}_{g,j} + \dot{J}_{g,j}$ ($j = O_2, CO, CO_2, NO, N_2O, N_2, HCN, CNO, NH_3, H_2O, SO_2, CH_4$)		$(d(\rho_j\varepsilon_{p,i}))/(dt) = \sum_{in} \dot{m}_j\varepsilon_{p,i} - \sum_{out} \dot{m}_j\varepsilon_{p,i} + \dot{R}_{s,j} + \dot{J}_{s,j}$ (j = char, limestone, bed material)	
Momentum Equation			
$(\partial(Cu\varepsilon_i))/(\partial t) + (\partial(Cu\varepsilon_i u))/(\partial r) = -(\partial(P\varepsilon_i))/(\partial r) - (\partial(\tau_{rr}\varepsilon_i))/(\partial r) - (\partial(\tau_{rz}\varepsilon_i))/(\partial z) - \beta(u - v)$		$(\partial(\rho v\varepsilon_{p,i}))/(\partial t) + (\partial(\rho v\varepsilon_{p,i}v))/(\partial r) = -(\partial(\tau_{rr}\varepsilon_{p,i}))/(\partial r) - (\partial(\tau_{rz}\varepsilon_{p,i}))/(\partial z) + \beta(u - v) - (\partial(G(\varepsilon)\varepsilon_{p,i}))/(\partial r)$	
$(\partial(Cu\varepsilon_i))/(\partial t) + (\partial(Cu\varepsilon_i u))/(\partial z) = -(\partial(P\varepsilon_i))/(\partial z) - (\partial(\tau_{zz}\varepsilon_i))/(\partial z) - (\partial(\tau_{zr}\varepsilon_i))/(\partial r) - \beta(u - v)$		$(\partial(\rho v\varepsilon_{p,i}))/(\partial t) + (\partial(\rho v\varepsilon_{p,i}v))/(\partial z) = -(\partial(\tau_{zz}\varepsilon_{p,i}))/(\partial z) - (\partial(\tau_{zr}\varepsilon_{p,i}))/(\partial r) + \beta(u - v) - (\partial(G(\varepsilon)\varepsilon_{p,i}))/(\partial z) + \rho_g\varepsilon_{p,i}$	
$\tau_{rr} = 2\mu(\partial u/\partial r) - {}^2/_3\mu((\partial u/\partial r) + (\partial u/\partial z))$	ideal gas equation	$\tau_{rr} = 2\ \mu(\partial v/\partial r) - {}^2/_3\mu((\partial v/\partial r) + (\partial v/\partial z))$	solids stress modulus
$\tau_{zz} = 2\mu(\partial u/\partial z) - {}^2/_3\mu((\partial u/\partial z) + (\partial u)/(\partial r))$	$C = (1/R_u)(P/T)$	$\tau_{zz} = 2\mu(\partial v/\partial z) - {}^2/_3\mu((\partial v/\partial z) + (\partial v/\partial r))$	$G(\varepsilon) = \partial\tau/(\partial(1 - \varepsilon)) = 10^{-8.76\varepsilon+5.43}$
$\tau_{rz} = \tau_{zr} = \mu((\partial u/\partial z) + (\partial u/\partial r))$		$\tau_{rz} = \tau_{zr} = \mu((\partial v/\partial z) + (\partial v/\partial r))$	
gas−solid friction coefficient			
$\beta = {}^3/_4C_D[(C\varepsilon_i(1 - \varepsilon_i))/(\varepsilon_i^{2.65})](1/d_p)\lvert u - v\rvert$		$C_D = (24/Re_p)(1 + 0.15Re_p^{0.687})$	$Re_p < 1000$
		$C_D = 0.44$	$Re_p \geq 1000$
Energy Equation			
$C\varepsilon_i\bar{c}_{p,g}(\partial T/\partial t) - uC\varepsilon_i\bar{c}_{p,g}(\partial T/\partial r) - uC\varepsilon_i\bar{c}_{p,g}(\partial T/\partial z) + \rho\varepsilon_{p,i}c_p(\partial T_p/\partial t) - u\rho\varepsilon_{p,i}c_p(\partial T_p/\partial r) - u\rho\varepsilon_{p,i}c_p(\partial T_p/\partial z) = R''' - \dot{Q}_{wall} + \mu\varepsilon_i\{2[(\partial u/\partial r)^2 + (\partial u/\partial z)^2] + {}^1/_3[(\partial u/\partial r) + (\partial u/\partial z)]^2\} + \mu\varepsilon_i\{2[(\partial v/\partial r)^2 + (\partial v/\partial z)^2] + {}^1/_3[(\partial v/\partial r) + (\partial v/\partial z)]^2\}$			

Numerical Solution. The model allows dividing the calculation domain into *mn* control volumes, in the radial and the axial directions and in the core and the annulus regions, respectively. In this study the calculation domain is divided into 8 × 50 control volumes in the radial and the axial directions and in the core and the annulus regions, respectively (Figure 1). With the cylindrical system of coordinates, a symmetry boundary condition is assumed at the column axis. The set of differential equations governing mass, momentum, and energy for the gas and solid phases are given in detail in a previous study[8] and are solved with a computer code developed by the author in FORTRAN language where the time step is 10^{-6} seconds (Table 2). The Gauss−Seidel iteration, which contains a successful relaxation method, and combined relaxation Newton−Raphson methods are used for solving procedure. Details about the solving procedure are given elsewhere.[8]

Inputs for the model are combustor dimensions and construction specifications (insulation thickness and materials), primary and secondary air flow rates, coal feed rate and particle size distribution, coal properties, Ca/S ratio, limestone particle size distribution, inlet pressure and temperature, ambient temperature, and the superficial velocity. The secondary air injection affects the concentration of oxygen, the bed voidage with increasing gas flow rate, the velocity profiles of the gas and the solid phases, and the overall bed temperature. A continuity condition is used for the gas phase at the top of the cyclone. The cyclone is considered to have 98% collection efficiency. The simulation model calculates the axial and radial distribution of voidage, velocity, particle size distribution, pressure drop, gas emissions, and temperature at each time interval for gas and solid phases both for dense bed and for riser. While investigating the effects of operational parameters, the mean bed temperature value is considered as 850 °C.

Comparison Data. The comparison data are obtained from three different size CFBC, which use different kinds of low-grade Turkish lignites, the 50 kW pilot scale CFBC using Beypazari lignite, the 80 kW pilot scale CFBC using Tuncbilek lignite, and industrial scale 160 MW CFBC using Can lignite (during the commissioning period). To test and validate the

Table 3. Proximate and Ultimate Analysis of Lignites

	Beypazari lignite	Can lignite	Tuncbilek lignite
proximate analysis (wt %)			
moisture	14.30	28.42	7.50
ash	35.20	14.34	23.70
volatile matter	31.50	29.74	27.50
fixed carbon	19.00	27.50	41.30
ultimate analysis (wt %, dry)			
C	37.20	40.85	59.29
H	2.90	4.01	4.61
N	1.30	2.25	2.10
O	14.90	12.79	11.54
S	2.60	8.40	1.81
ash	41.10	31.70	20.65
LHV (MJ/kg, dry)	10.283	11.704	22.062

Table 4. Operating Parameters of the Experimental Data Referred to in This Study

operating parameters	50 kW pilot scale combustor[2]	80 kW pilot scale combustor[28]	160 MW industrial scale combustor[13]
coal feed rate	15.1 kg/h	6−7.7 kg/h	110−120 t/h
operation velocity	1.75 m/s	3.60−9.23 m/s	4−6 m/s
bed temperature	850−900 °C	860−900 °C	850−900 °C
primary/secondary air ratio	2/3		2/3
size of coal feed	0.03−0.90 mm		0.1−9.0 mm
mean size of sorbent feed	0.071−0.100 mm		0.1−0.15 mm

model presented in this paper, the same input variables in the tests are used as the simulation program input in the comparisons.

In the pilot-scale CFBC of 50 kW, the riser is a cylinder of 12.5 cm i.d. and 130 cm combustor height.[2] Beypazari lignite is fed to the combustor, and its properties are shown in Table 3. Limestone is used as adsorbent. In the experiments 20% excess air is used. A more detailed description of the experiment is given in Ozkan and Dogu.[2] The considered parameters and computation conditions are given in Table 4.

In the pilot-scale CFBC of 80 kW the riser is a cylinder of 12.5 cm i.d. and has 180 cm combustor height.[28] Tuncbilek lignite is fed to the combustor, and its properties are shown in Table 3. Limestone is used as adsorbent. Silica sand and ash were used as bed materials. The weighted average particle sizes are determined to be 56 μm for sand particles. A more detailed description of the experiment is given in Topal et al.[29] The considered parameters and computation conditions are given in Table 4.

The industrial-scale CFBC of 160 MW (Can Power Plant) has a combustor of 700 cm × 1400 cm square cross-section and 3700 cm height.[13] The combustor has a square cross-section, but the lower section has less cross-sectional area than the upper section. The technical parameters of the CFBC are steam capacity of 485 t/h, and superheated steam temperature and pressure of 543 °C and 17.5 MPa, respectively. The secondary air ports are located at 500 cm from the distributor. The design fuel for the bed is Can lignite, which is fed to the combustor, and its properties are shown in Table 3. Limestone is used as adsorbent. The operating parameters of data used for the comparison of the CFB model is shown in Table 4.

Results and Discussion

The emission of NO_x depends on temperature because the NO_x oxidation changes significantly with temperatures.[25] The

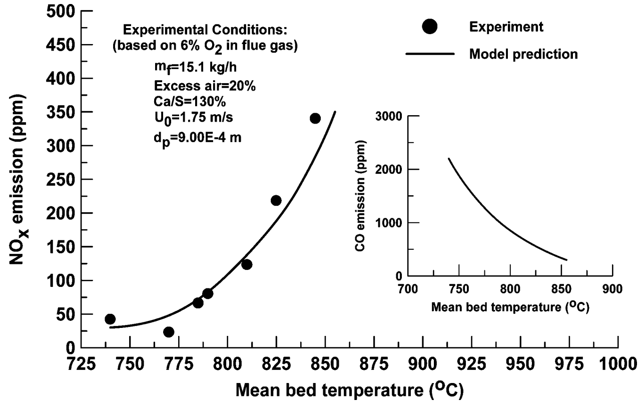

Figure 2. Comparison of model NO_x emission predictions with experimental data for 50 kW pilot scale CFBC[2] with regard to the mean bed temperature.

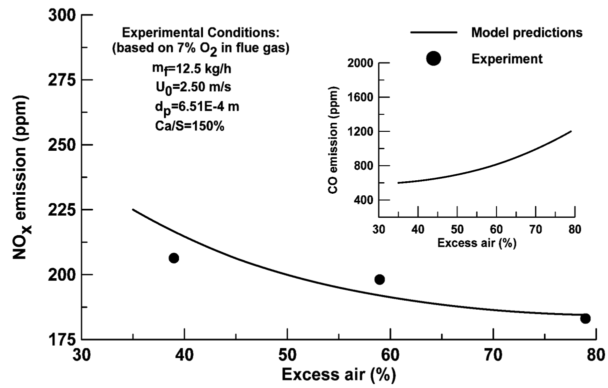

Figure 3. Comparison of model NO_x emission predictions with experimental data for 80 kW pilot-scale CFBC[28] with regard to the excess ratio.

increase of NO_x emissions with combustor temperature is observed in Figure 2 for 50 kW pilot scale CFBC, whereas below 800 °C NO_x emissions are rather low. Over 800 °C some increase in NO_x emissions is observed. The carbon content depends on temperature and the combustion rate of char, which could be determined from the kinetic model of char combustion. With the increase of temperature, the reaction rate constant of char combustion increases, which leads to lower char concentration in the combustor and lower CO concentration. The model CO concentration profile is also shown in Figure 2. The reduction of NO_x emissions is proportional to the presence of char particles and CO concentration in the control volume. Due to the lack of enough char particles and lower CO concentration, the reduction of NO_x to N_2 decreases. The results of Lin et al.[30]'s study also verify this phenomena. It is clearly seen from Figure 2 that both experimental data and model predictions show the close agreement.

In Figure 3, NO_x emissions based on 7% O_2 in the flue gas for 80 kW pilot-scale CFBC are plotted with respect to excess air, which rages between 35 and 80%, where the mean bed temperature range is 857−768 °C (for 857 °C, the excess air is 35%; for 768 °C, the excess air is 80%). The model CO concentration profile is also shown in Figure 3. In Figure 3, the NO_x emission decreases with increasing excess air as observed in both experimental data and model predictions. Although the amount of oxygen increases with increasing excess air, decreasing bed temperature causes a negative effect on coal combustion

(28) Topal, H. *Experimental investigation of the hydrodynamic, combustion and emission properties of a circulating fluidized bed*; Ph.D. Thesis, Gazi University Institute of Science and Technology: Turkey, 1999.
(29) Topal, H.; Atimtay, A. T.; Durmaz, A. *Fuel* **2004**, *82*, 1049–1056.

(30) Lin, C. H.; Teng, J. T.; Chyang, C. S. *Combust. Flame* **1997**, *110*, 163–172.

Table 5. Comparison of Simulation Results with 160 MW Industrial CFBC Test Results[13]

		T (°C)			NO_x (ppm)		
time (min.)	coal feed (t/h)	model	data	error (%)	model	data	error (%)
30	119.1	798.50	807.1	1.06	73.425	72.825	0.82
60	119.0	798.79	809.1	1.27	72.675	71.925	1.03
90	116.9	800.36	812.4	1.48	73.17	74.025	1.14
120	116.3	798.59	814.9	2.00	69.39	69.525	0.18
150	116.0	798.40	812.3	1.71	76.5225	76.875	0.45
180	118.4	798.26	805.5	0.89	74.2125	74.025	0.29
210	113.8	804.01	809.3	0.65	74.295	73.65	0.87

efficiency, which results in lower levels of NO_x formation.[31] Decreasing combustion efficiency also causes higher carbon content in the combustor. Thus, the reduction rate of NO_x increases. Another explanation of decreasing NO_x emission is the gas dilution caused by increasing excess air. As for CO concentrations, decreasing bed temperature results in the reaction rate of char combustion to decrease and also a decrease of the CO oxidation rate constant, which leads to higher CO emission, which is another reason for lower NO_x emission as displayed in Figure 3. This phenomenon is also observed in the studies of Desroches-Ducarne et al.[32]

For the 160 MW industrial-scale CFBC, temperature and NO_x emissions response in flue gases simulation and test results at the riser exit are compared at different coal feed rates, and the results are presented in Table 5. It is seen that the simulation results are in good agreement with industrial-scale CFBC data as well.

Model predictions are in good agreement with both industrial and small-scale CFBCs, which is an indication that the model is flexible enough to be used in different CFB applications and simulates under a wide range of operating conditions such as coal type, combustor temperature, and excess air ratio. Moreover, both experimental data and model predictions show the close agreement and have low and acceptable levels of NO_x emissions.

Effects of Operational Parameters. In the present study, the variations of the overall NO_x emission under different operational conditions such as excess air (20−100%), bed operational velocity (4.15−6.50 m/s), and coal particle diameter (540−852 μm) are analyzed for the 80 kW pilot-scale CFBC conditions with the developed and validated 2D model with respect to these emissions.

Excess air is one of the operating variables affecting both thermal and environmental performances of a CFB. In general, the carbon combustion efficiency increases with increasing excess air. This indicates that air staging could improve combustion. An increase of excess air gives an increase in the mean oxygen concentration in the bed. This causes the combustion rate of char to increase by the amount of oxygen present, thus increasing the carbon combustion efficiency. Since the size of the CFBC considered in this study is quite small, a decrease of the combustion efficiency is observed due to the combustion losses increase with increasing excess air. Excess air affects combustion efficiency in two ways: one is due to higher heat losses with increasing flue gas flow rates to the ambient. As expected, decreasing the temperature decreases the carbon combustion efficiency due to the decrease in the reaction rates. The other one is bed temperature decreases as the excess air increases, and it affects the carbon combustion efficiency due to decreases in the reaction rates and as a result higher carbon content in the mass discharged from the combustor. This phenomenon is also observed in the study of Bhatt.[33] As mentioned above, decreasing bed temperature generates a decrease of the CO oxidation rate constant.

The reduction of NO_x emissions is proportional to the presence of char particles and CO concentration in the control volume. Higher carbon content and CO concentration lead to lower NO_x emissions as clearly seen in Figure 4. Another explanation of decreasing NO_x emission is the gas dilution caused by increasing excess air, which is also observed in the studies of Liu and Gibbs[34] and Xie et al.[35] A smaller mean size of char in the combustor will result in a lower emission of NO_x if other parameters are kept unchanged,[36] as clearly seen from Figures 4 and 5.

The bed operational velocity in the combustor is one of the basic design variables of the process. The reason is that with the increase of bed operating velocity the hydrodynamic condition of the combustor changes. The bed operational velocity increases circulation flow rates of solids and results in a decrease of the mean residence time of char particles in the combustor. This situation causes insufficient residence time for complete combustion and, consequently, CO emission increases and the mean bed temperature decreases. On the other hand, suspension density in the bed decreases with increasing superficial velocity, which results in lower level of char particles in the control volume. Therefore, NO_x emissions increase with the superficial velocity of the combustor (Figure 5). This phenomenon is also observed in the study of Scala and Salatino.[37] The high fuel-N contents of the large size of particles causes the

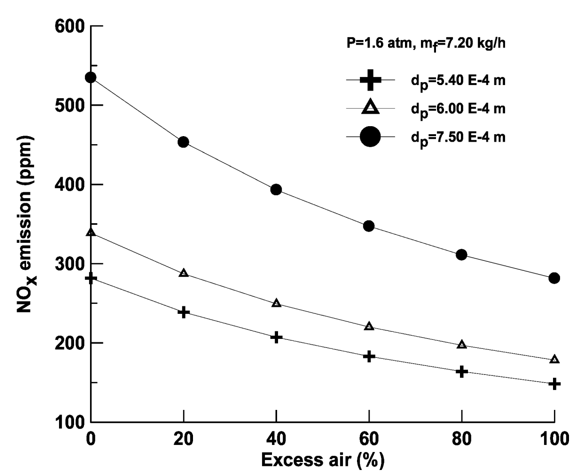

Figure 4. Effect of excess air ratio on the overall NO_x emission from the combustor. T_{mean} = 868 °C (for the excess air = 0%); T_{mean} = 756 °C (for the excess air = 100%).

(31) Madhiyanon, T.; Lapirattanakun, A.; Sathitruangsak, P.; Soponronnarit, S. *Combust. Flame* **2006**, *146* (1−2), 232–245.
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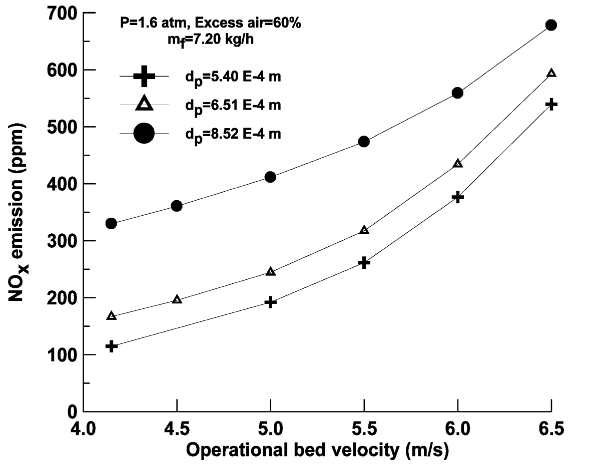

Figure 5. Effect of operational bed velocity on the overall NO_x emission from the combustor.

high rates of NO_x emission formation, as is clearly seen from Figures 4 and 5.

Conclusions

The variations of the overall NO_x emission under different operational conditions such as excess air (20−100%), bed operational velocity (4.15−6.50 m/s), and coal particle diameter (540−852 μm) are analyzed with the developed and validated 2D model with respect to these emissions. As a result of this investigation, the general tendency is for a decrease in the NO_x emission as excess air increases. NO_x emission increases with the operational bed velocity. The present study proves that CFB combustion allows clean and efficient combustion of low grade coal which is demonstrated by the fact that both experimental data and model predictions have low and acceptable levels of NO_x emission.

Appendix

Nomenclature

C = gas concentration (kmol/m^3)
Ca/S = calcium to sulfur ratio (−)
C_D = drag coefficient (−)
c_p = specific heat of solids (kJ/kg K)
$\bar{c}_{p,g}$ = specific heat of gas (kJ/kmol K)
D_g = diffusivity coefficient for oxygen in nitrogen (m^2/s)
d_p = particle diameter (m)
G = solid stress modulus (N/m^2)
$\dot{J}$ = mass transfer rate via density difference per unit volume (kmol/m^3 s)
k = rate constant (m/s)
k_c = char combustion reaction rate (kg/s)
k_{cr} = kinetic rate (1/s)
k_{cd} = diffusion rate (1/s)
k_g = gas conduction heat transfer coefficient (W/m/K)
M_c = molecular weight of carbon (kg/mol)
$\dot{m}$ = mass flow rate (kg/s)
m_f = feed rate (kg/h)
$\dot{n}$ = gas flow rate (mol/s)
P = pressure (Pa)
$\dot{Q}$ = heat flux (W)
R = reaction rate (mol/ s), (mol/cm^3 s)
R_g = gas constant (kJ/mol/K)
$\dot{R}$ = mass flow rate generated/consumed from chemical processes per unit volume (kmol/m^3 s)
R''' = total heat generated/consumed per unit volume (W/m^3)
Re = Reynolds number (−)
R_u = Universal gas constant (kJ/kmol K)
Sc = Schmidt number (−)
T = temperature (K)
T_{mean} = mean bed temperature (K)
T_p = particle temperature (K)
U_0 = superficial velocity (m/s)
u = gas velocity (m/s)
v = particle velocity (m/s)

Greek Letters

β = gas−solid friction coefficient
ε = void fraction
ε_p = solids volume fraction
Φ = mechanism factor
μ = gas viscosity (kg/m s)
ρ = particle density (kg/m^3)
τ = shear stress (N/m^2)

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